Pastry: routing hops versus network size

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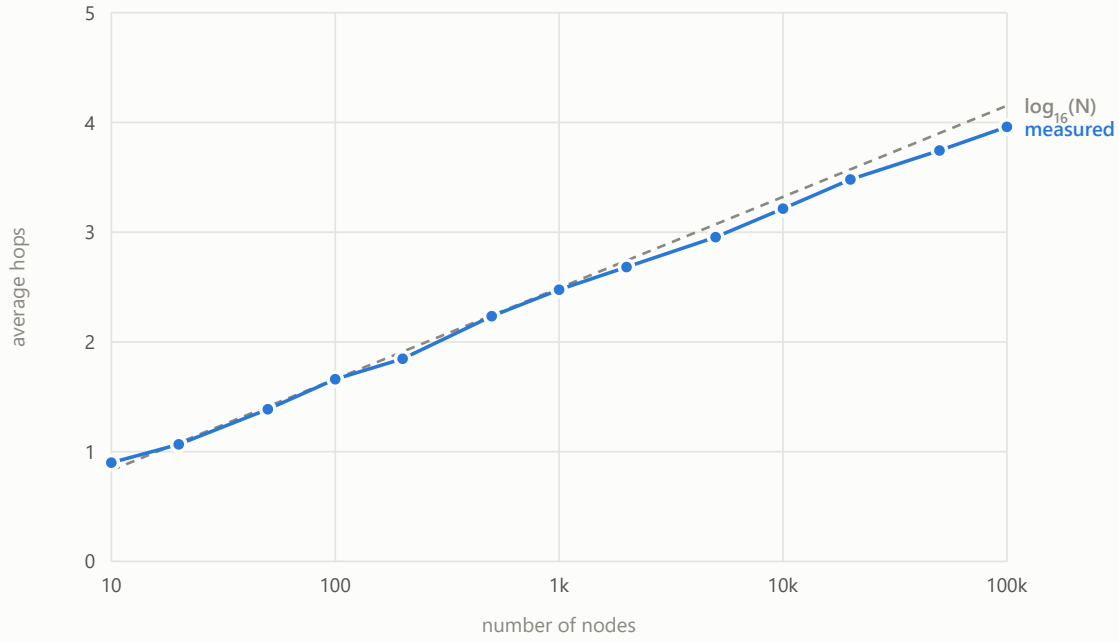
Each peer is one Erlang actor. Every peer makes two lookups for randomly chosen stored objects, and the simulator reports how many node-to-node hops each lookup travelled before it was delivered. Networks of up to 5,000 nodes are averaged over three independent runs; larger ones are single runs.

Average hops

measured $\log_{16}(N)$

Average hops per lookup

Network size on a logarithmic axis. Pastry's bound is $\lceil \log_{16} N \rceil$; the dashed line is $\log_{16} N$ itself.



What the measurements show

Routing is logarithmic, as Pastry promises. On a logarithmic x-axis the measured curve is a straight line, which is exactly what O(log N) looks like. Growing the network ten-thousand-fold, from 10 to 100,000 nodes, adds only about three hops.

It runs slightly under $\log_{16} N$. The average sits at or below the dashed line at every size from 100 nodes upward. That is expected rather than lucky: $\log_{16} N$ is the number of hops needed to resolve a key one hex digit at a time, but the leaf set lets the last few digits be resolved in a single hop, and the Pastry paper reports the same effect.

Every lookup found its object. 413,280 of 413,280 lookups across all runs. Separately, routing was checked against an independent oracle — the true numerically-closest node, found by brute force over every node id rather than taken from any node's own tables — and every probe reached it.

Largest network

100,000 nodes: 200,000 of 200,000 lookups found, averaging 3.96 hops against $\log_{16}N = 4.15$. On an 8 GB machine the limit is the time to join peers one at a time, not memory.

Measurements

Nodes	Average hops	$\log_{16}N$	Found
10	0.900	0.830	60 / 60
20	1.067	1.080	120 / 120
50	1.387	1.411	300 / 300
100	1.660	1.661	600 / 600
200	1.847	1.911	1,200 / 1,200
500	2.235	2.241	3,000 / 3,000
1,000	2.476	2.491	6,000 / 6,000
2,000	2.682	2.741	12,000 / 12,000
5,000	2.955	3.072	30,000 / 30,000
10,000	3.214	3.322	20,000 / 20,000
20,000	3.480	3.572	40,000 / 40,000
50,000	3.744	3.902	100,000 / 100,000
100,000	3.960	4.152	200,000 / 200,000